

YONGGANG JIANG

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Max Planck Institute for Informatics (MPI-INF), E14 Room 315, Saarland Informatics Campus

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EDUCATION

Max Planck Institute for Informatics and Saarland University PhD in Computer Science Advisor: Danupon Nanongkai	<i>Apr. 2023 – Present</i>
Max Planck Institute for Informatics and Saarland University Preparatory Phase for Doctoral Study	<i>Oct. 2021 – Mar. 2023</i>
Nanjing University B.S. in Computer Science and Technology GPA: 94.4/100, Ranking: 1/31	<i>Sep. 2017 – Jul. 2021</i>
University of California, Berkeley Berkeley International Study Program GPA: 4.0/4.0	<i>Jan. 2020 – May 2020</i>

HONORS & AWARDS

Distinguished Paper Award, SPAA 2025	<i>2025</i>
Outstanding Graduate, Nanjing University	<i>2021</i>
National Elite Program Scholarship, Outstanding Prize (top 1)	<i>2018, 2019, 2020</i>
China Collegiate Programming Contest (CCPC), Gold Medal	<i>2018</i>
National Olympiad in Informatics in Provinces (NOIP), First Prize	<i>2015</i>

PUBLICATIONS

*Co-authors are listed in alphabetical order.

- [1] Reviving Thorup's Shortcut Conjecture
Aaron Bernstein, Bernhard Haeupler, Gary Hoppenworth, Yonggang Jiang, Maximilian Probst Gutenberg, Seth Pettie, Thatchaphol Saranurak, Leon Schiller.
In submission.
- [2] Minimum s-t Cuts with Fewer Cut Queries
Danupon Nanongkai, Yonggang Jiang, Pachara Sawettamalya.
In submission.
- [3] Reducing Shortcut and Hopset Constructions to Shallow Graphs
Bernhard Haeupler, Yonggang Jiang, Thatchaphol Saranurak.
In submission.
- [4] Directed and Undirected Vertex Connectivity Problems are Equivalent for Dense Graphs
Yonggang Jiang, Sagnik Mukhopadhyay, Sorrachai Yingchareonthawornchai.
In submission.
- [5] Parallel Small Vertex Connectivity in Near Linear Work and Polylogarithmic Depth
Yonggang Jiang, Changki Yun.
In submission.

- [6] Shortcuts and Transitive-Closure Spanners Approximation
Parinya Chalermsook, Yonggang Jiang, Sagnik Mukhopadhyay, Danupon Nanongkai.
In submission.
- [7] New Oracles and Labeling Schemes for Vertex Cut Queries
Yonggang Jiang, Merav Parter, Asaf Petruschka.
In submission.
- [8] Perfect Simulation of Las Vegas Algorithms via Local Computation
Xinyu Fu, Yonggang Jiang, Yitong Yin.
In submission.
- [9] Parallel $(1 + \epsilon)$ -Approximate Multi-Commodity Mincost Flow in Almost Optimal Depth and Work
Bernhard Haeupler, Yonggang Jiang, Yaowei Long, Thatchaphol Saranurak, Shengzhe Wang.
IEEE Symposium on Foundations of Computer Science (FOCS) 2025.
- [10] Parallel Minimum Cost Flow in Near-Linear Work and Square Root Depth for Dense Instances
Jan van den Brand, Hossein Gholizadeh, Yonggang Jiang, Tijn de Vos.
ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2025.
Distinguished Paper Award
- [11] Deterministic Vertex Connectivity via Common-Neighborhood Clustering and Pseudorandomness
Chaitanya Nalam, Yonggang Jiang, Thatchaphol Saranurak, Sorrachai Yingchareonthawornchai.
ACM Symposium on Theory of Computing (STOC) 2025.
- [12] Global vs. s-t Vertex Connectivity Beyond Sequential: Almost-Perfect Reductions & Near-Optimal Separations
Joakim Blikstad, Yonggang Jiang, Sagnik Mukhopadhyay, Sorrachai Yingchareonthawornchai.
ACM Symposium on Theory of Computing (STOC) 2025.
- [13] Parallel and Distributed Exact Single-Source Shortest Paths with Negative Edge Weights
Vikrant Ashvinkumar, Aaron Bernstein, Nairen Cao, Christoph Grunau, Bernhard Haeupler, Yonggang Jiang, Danupon Nanongkai, Hsin Hao Su.
European Symposium on Algorithms (ESA) 2024.
- [14] Finding a Small Vertex Cut on Distributed Networks
Yonggang Jiang, Sagnik Mukhopadhyay.
ACM Symposium on Theory of Computing (STOC) 2023.
- [15] Robust and Optimal Contention Resolution without Collision Detection
Yonggang Jiang, Chaodong Zheng.
ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2022.
- [16] Tight Trade-off in Contention Resolution without Collision Detection
Haimin Chen, Yonggang Jiang, Chaodong Zheng.
ACM Symposium on Principles of Distributed Computing (PODC) 2021.

RESEARCH VISITS & SEMINAR TALKS

Carnegie Mellon University, Pittsburgh, PA, USA
Hosted by Jason Li.

Aug. 2025

Talk: Parallel $(1 + \epsilon)$ -Approximate Multi-Commodity Mincost Flow in Almost Optimal Depth and Work

University of Michigan, Ann Arbor, MI, USA

July. 2025

Hosted by Thatchaphol Saranurak.

ETH Zurich, Zurich, Switzerland

Feb. 2025

Hosted by Sorrachai Yingchareonthawornchai.

Tsinghua University, Beijing, China

Dec. 2024

Hosted by Ran Duan.

Talk: New Tools and Faster Algorithms for Vertex Min-Cut

Nanjing University, Nanjing, China

Dec. 2024

Hosted by Yitong Yin.

Talk: New Tools and Faster Algorithms for Vertex Min-Cut

Shanghai University of Finance and Economics, Shanghai, China

Dec. 2024

Hosted by Pinyan Lu.

Talk: New Tools and Faster Algorithms for Vertex Min-Cut

ETH Zurich, Zurich, Switzerland

Nov. 2024

Hosted by Weiming Feng.

Talk: New Tools and Faster Algorithms for Vertex Min-Cut

INSAIT, Sofia, Bulgaria

Sep. 2024

Hosted by Bernhard Haeupler.

University of Warwick, Coventry, UK

Jun. 2024

Hosted by Sayan Bhattacharya.

Talk: Deterministic k -Vertex Connectivity in k Max-Flows

University of Sheffield, Sheffield, UK

Jun. 2024

Hosted by Sagnik Mukhopadhyay.

Nanjing University, Nanjing, China

Nov. 2023

Hosted by Penghui Yao.

Rutgers University, New Brunswick, NJ, USA

Jun. 2023

Hosted by Aaron Bernstein.

Aalto University, Helsinki, Finland

Apr. 2023

Hosted by Parinya Chalermsook.

Aalto University, Helsinki, Finland

Dec. 2022

Hosted by Parinya Chalermsook.

University of Sheffield, Sheffield, UK

May 2022

Hosted by Sagnik Mukhopadhyay.

CONFERENCE & WORKSHOP TALKS

Parallel Minimum Cost Flow in Near-Linear Work and Square Root Depth for Dense Instances

SPAA 2025, Portland, OR, USA

July. 2025

Global vs. s - t Vertex Connectivity Beyond Sequential: Almost-Perfect Reductions & Near-Optimal Separations

STOC 2025, Prague, Czech Republic

Jun. 2025

Contributed Talk: Global vs. s-t Vertex Connectivity Beyond Sequential: Almost-Perfect Reductions & Near-Optimal Separations
HALG 2025, Zurich, Switzerland Jun. 2025

Contributed Talk: Deterministic k-Vertex Connectivity in k Max-Flows
HALG 2024, Warsaw, Poland Jun. 2024

Parallel, Distributed, and Quantum Exact Single-Source Shortest Paths with Negative Edge Weights
ESA 2024, London, UK Jun. 2024

Finding a Small Vertex Cut on Distributed Networks
STOC 2023, Orlando, FL, USA Jun. 2023

Robust and Optimal Contention Resolution without Collision Detection
SPAA 2022, online Jul. 2022

Tight Trade-off in Contention Resolution without Collision Detection
PODC 2021, online Jul. 2021

TEACHING EXPERIENCE

Seminar: Foundations of Machine Learning, MPI-INF *Summer 2023*
 Role: Coordinator and Speaker.

Data Structures and Algorithms, Nanjing University *Fall 2020, Fall 2019*
 Role: Teaching Assistant.

STUDENTS & MENTORING

Changki Yun (B.Sc.) *Fall 2024*
 Internship at MPI-INF, from SNU, Korea.
 Outcome: Paper on parallel vertex connectivity (in submission)

Hossein Gholizadeh (B.Sc.) *Summer 2024*
 Bachelor's Thesis Internship at MPI-INF, from KIT, Germany.
 Outcome: Paper in SPAA 2025 (Distinguished Paper Award)

Alireza Danaei (B.Sc.) *Summer 2023*
 Internship at MPI-INF, from Sharif Univ., Iran

Vassilis Xanthopoulos (M.Sc.) *Spring 2023*
 Master's Thesis Internship at MPI-INF, from NTUA, Greece